STATISTICAL LEARNING and LARGE DATA

# Disentangling Mutagenesis from Misdiagnosis in Secondary Lymphomas

*An Integrated Macro-Epidemiological and Transcriptomic Framework*

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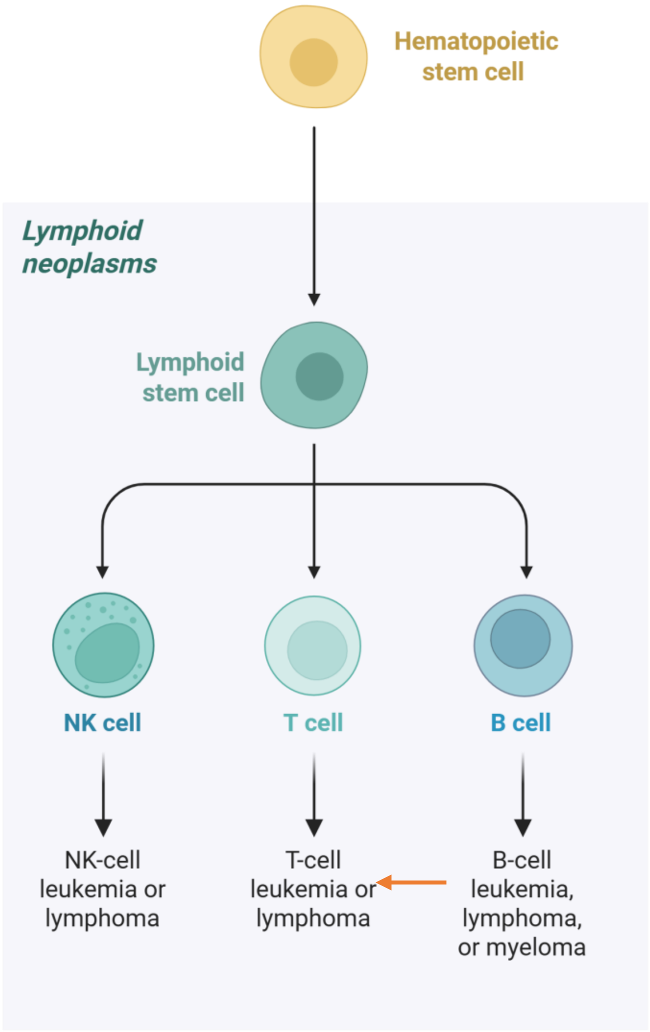
# Introduction: When Morphology Is Not Enough




**Classical Hodgkin Lymphoma (cHL):**
B-cell derived malignancy:
- Approximatly 10% of lymphomas
- Cure rates axceeding 80% in advance disease.

↓

**Secondary non-Hodgkin Lymphoma (sNHL):**
- Aggressive behaviour
- Frequent chemoresistance
- 5 year overall survival is around 35-44%

# Non-Mutually-Exclusive Mechanisms

sNHL may arise through non-mutually exclusive mecahnisms

## 1 Therapy-associated

Mutagenic effects of cytotoxic agents and radiation: historically the dominant explanatory frame.

## 2 Diagnostic overlap

A different lymphoma was already present but was not recognised at the initial biopsy.

***→ OUR GOAL: quantify the relative contribution of these factors***

## 3 Baseline heterogeneity

Patients with severe initial disease carry intrinsic biological vulnerability beyond treatment.

## 4 Shared clonal precursor

Both lesions could arise from a common pre-malignant clone, e.g. via TET2/DNMT3A mutations.

***→ DUAL SCALE framework:***

→ **Macro-Epidemiology**: A rigorous competing-risk survival architecture on 45,530 SEER patients.
→ **Micro-Genomics**: A penalized, high-dimensional transcriptomic pipeline to extract a stable molecular signature of sNHL.

# Macro-Epidemiology

Evaluating longitudinal sNHL risk involves a fundamental mathematical challenge: treated HL patients face a substantial risk of non-cancer-related mortality. We address this by employing two distinct model classes for two distinct questions.

## Fine-Gray Sub-distribution Hazard Model:

- ***Mechanism***: Computes the sub-distribution hazard γ(t) by conditioning on:

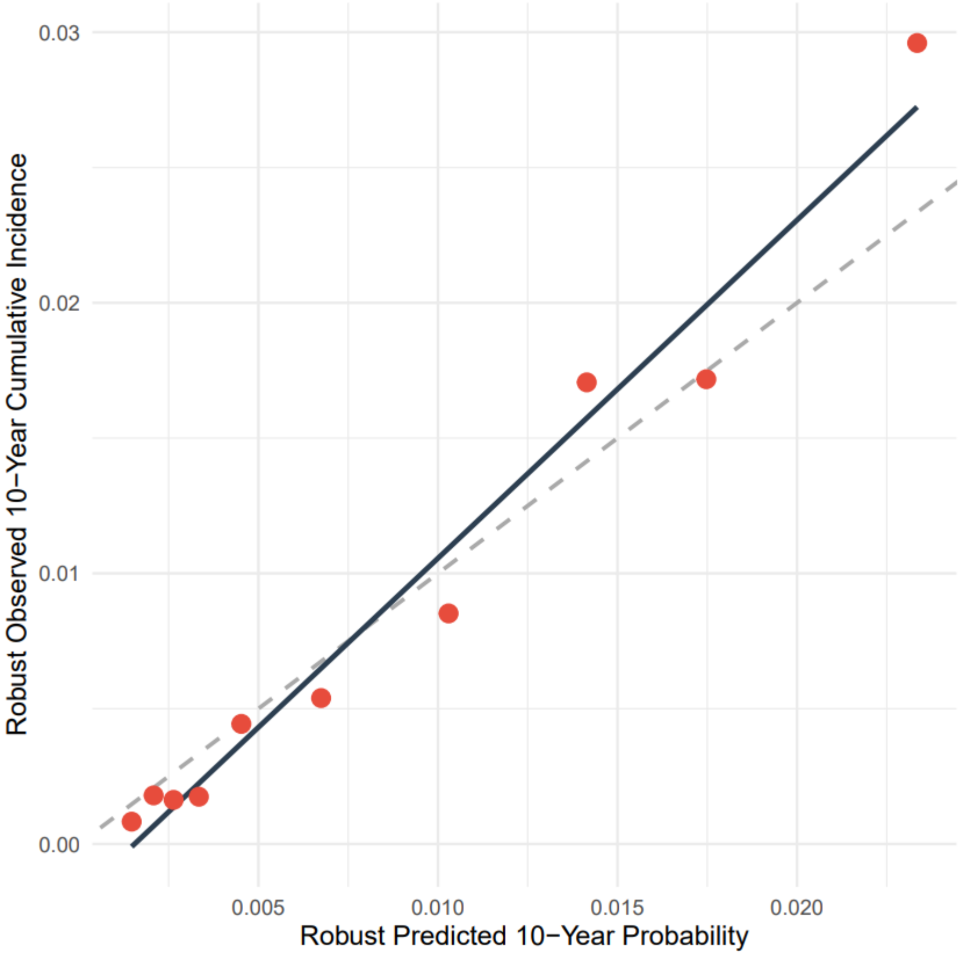
$$T > t \cup (T \leq t \cap J \neq 1)$$

- ***Why chosen***: rather than artificially removing deceased patients, it retains them in the risk set with an asymptotic zero-risk weight. This provides a real-word, unbiased estimation of the 10-year Cumulative Incidence Function (CIF).

## Cause-Specific Cox Model (For Etiological Inference) Mechanism:

- Models the cause-specific hazard hcs(t) strictly for the event of interest. Why chosen: Unlike Fine-Gray, this treats competing deaths strictly as independent censoring.
- This isolates the pure biological mutagenic hazard of sNHL development, allowing us to evaluate if chemotherapy intrinsically promotes lymphomagenesis in patients who remain alive and at risk.

# Fine-Grey Prediction and Validation

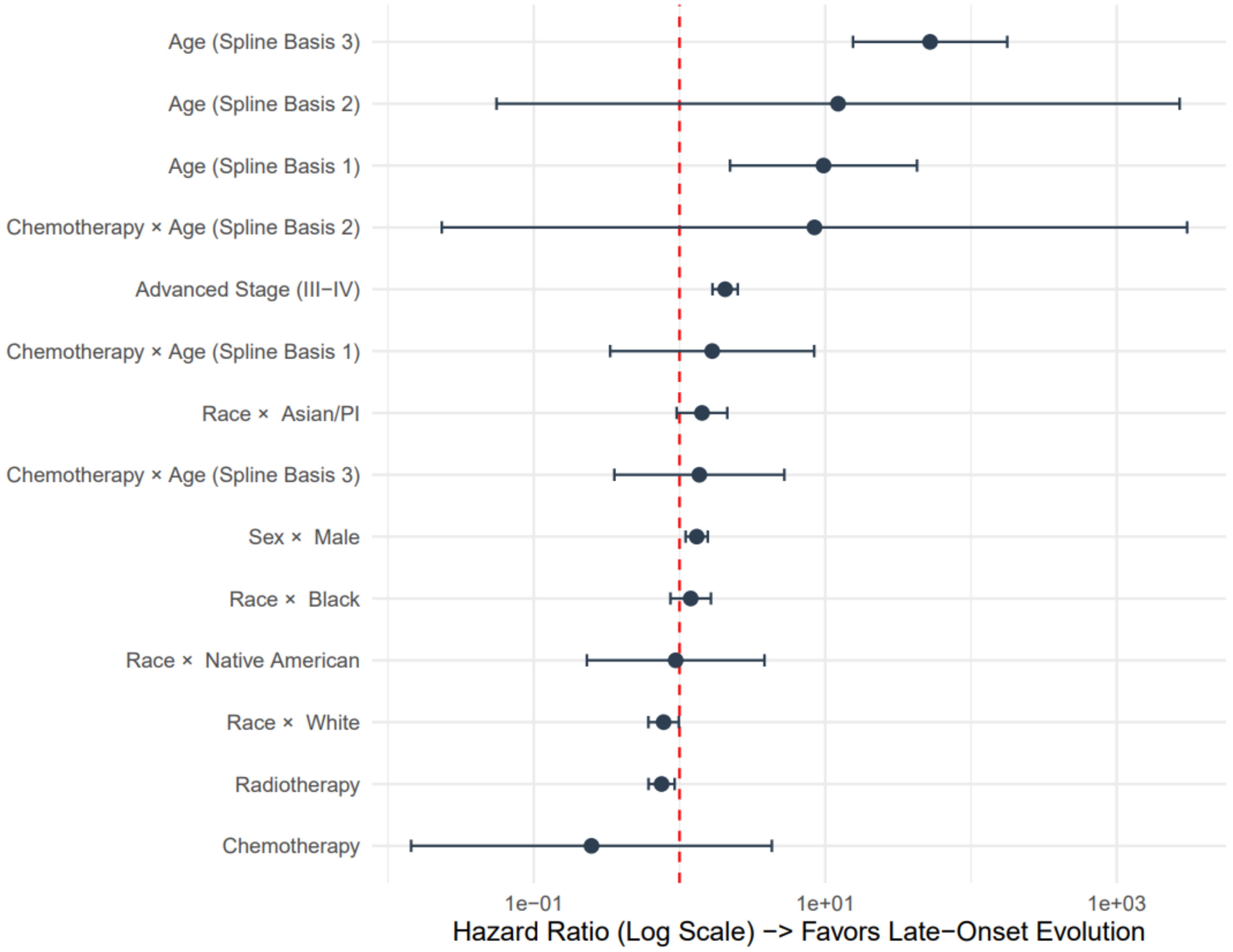



The model shows excellent 10-year calibration: predicted and observed probabilities are closely aligned across all risk groups, even under missing-data uncertainty.

# Cause Specific Etiology

- Systemic chemotherapy was not an independent significant risk factor for sNHL *(p = 0.339)*
- The strongest predictors were pre-existing clinical vulnerabilities, especially: Advanced stage: csHR = 2.06, $p < 0.001$ Age-related non-linear risk dynamics
- Radiotherapy showed only a weak negative association, not supporting a major mutagenic effect.

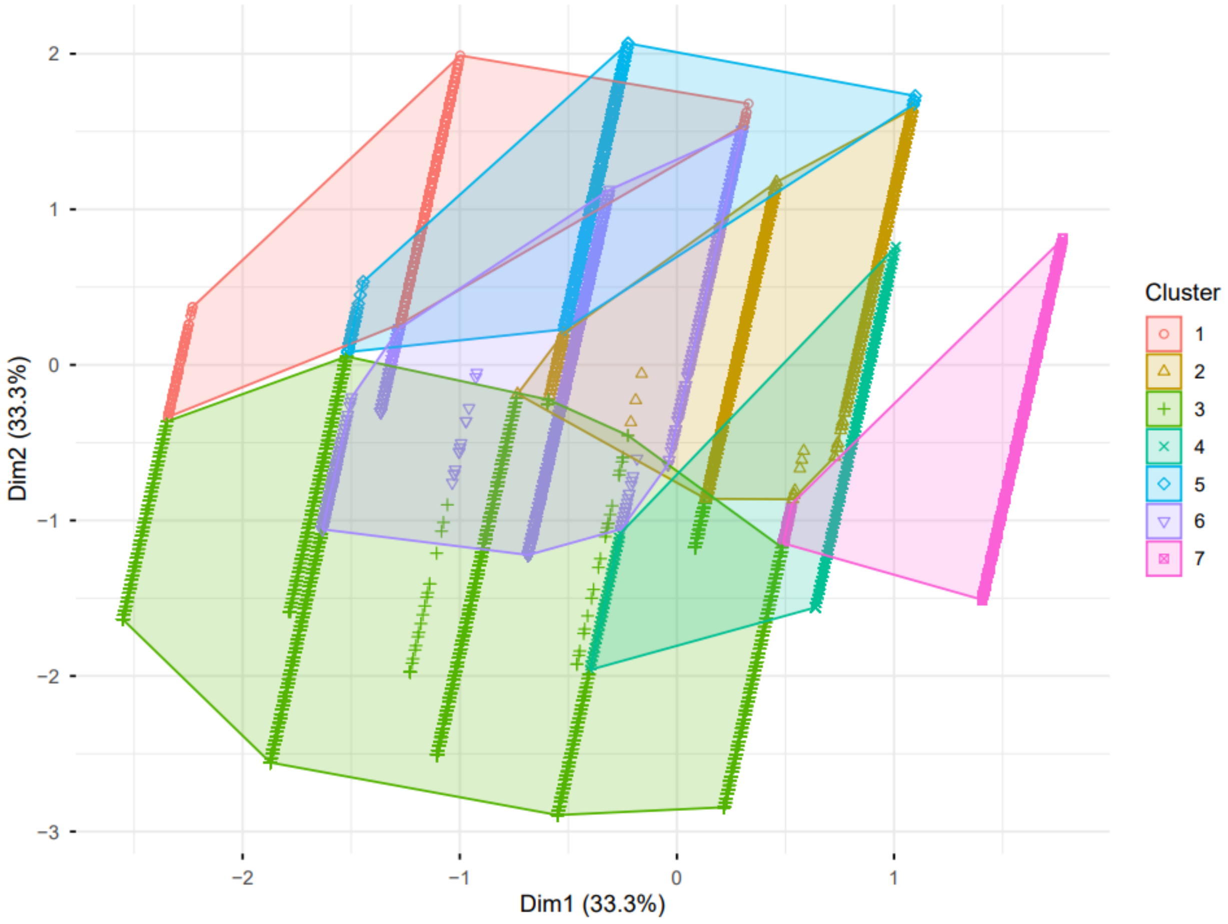
**Overall**, the data suggest that early sNHL is more consistent with baseline disease severity / diagnostic or biological predisposition than with delayed treatment-induced mutagenesis.



# Macro Methods

**Unsupervised baseline vulnerability mapping:**

- **MICE + Nelson–Aalen estimator** preserved survival/time-to-event structure during imputation.
- **FAMD** combined continuous and categorical clinical variables.
- First **3 components explained 71%** of variance.
- **K-Medoids selected K = 7** clusters using silhouette statistics and model complexity.
- Patients formed a **dense clinical continuum**, not rigid isolated groups.




**Take-home:** sNHL vulnerability reflects subtle baseline variation across age, stage, and severity rather than simple treatment-defined categories.

# Micro Methods I (GSE19069 T-cell NHL)

Transcriptomic features ($p \approx 54,000$) vastly exceed samples ($n \approx 150$), in a $n \ll p$ setting.

- Missing probes were inferred via a **distance-weighted** $k$-**Nearest Neighbors** ($k$-NN) algorithm. Crucially, the $k$-NN manifold was computed **exclusively within the training set boundary** to definitively prevent data leakage.
- We employed **SIS** to reduce dimensions by retaining the $d = \lfloor n/\log(n) \rfloor$ features with the largest marginal correlation $\boldsymbol{\omega} = \mathbf{X}^T\mathbf{y}$ with the phenotype. This was embedded inside $B = 100$ subsampling iterations, enforcing a $\geq 50\%$ selection threshold to resolve the instability of SIS to data perturbations.

## Sure Screening Property

SIS possesses the mathematically proven *sure screening property*:

$$\mathbb{P}(M^* \subseteq M_d) \underset{n \to +\infty}{\longrightarrow} 1,$$

where $M^*$ is the true sparse model.

# Micro Methods II

- **Elastic Net Regularization** minimizes the objective function by balancing Lasso ($L_1$-norm) and Ridge ($L_2$) penalties:
$$\beta = \arg\min_{\boldsymbol{\beta}} \left( \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2 \right).$$
While Lasso generates perfect sparsity but arbitrarily drops highly correlated genes, adding the Ridge penalty ($\alpha = 0.5$) retains entire functional groups of correlated biological pathways, essential for genomics.
- In transcriptomics, we are rarely interested in the isolated expression of single genes. Instead, the goal is to capture shifts in entire biological complexes and microenvironmental networks moving in concert. This is why we utilized **GSEA**, which computes a threshold-free running-sum statistic across the entire ranked transcriptome, making it preferable to standard Over-Representation Analysis (ORA), which applies arbitrary statistical cutoffs and aggressively discards moderate, yet highly coordinated, biological signals.

# 22-Gene signature

The Nested SIS pipeline successfully isolated a **22-gene signature**.

Specifically, **C9orf47** emerged with "Ultra Stability" ($\geq$ 90% selection frequency), flagging it as a crucial transcriptomic anchor.

Furthermore, **IL-21** achieved "High Stability" (81% frequency). It is the master cytokine defining the T-follicular helper (Tfh) lineage. Its pathological overexpression is the primary driver of B-cell activation and vascular proliferation, robustly validating the biological integrity of the signature.

# AITL vs. PTCL-NOS

Training an algorithm to directly separate classical HL from NHL would trivially extract lineage markers (e.g., Reed-Sternberg features vs. T-cell markers), yielding a 100% AUC, but completely failing to test the misdiagnosis hypothesis.

## AITL as a Mimicker

AITL harbors few neoplastic T-cells but induces a 90% reactive inflammatory infiltrate (eosinophils, plasma cells, EBV+ B-blasts) making it morphologically indistinguishable from classical HL.

This is why we use PTCL-NOS (a T-cell lymphoma lacking AITL's deceptive microenvironment) as a contrast background. This forces the Elastic Net to extract the pure AITL transcriptomic signature. Then, we project an external cohort of officially diagnosed HL patients into this newly defined AITL mathematical space. If standard pathology were infallible, HL would form an isolated cluster.

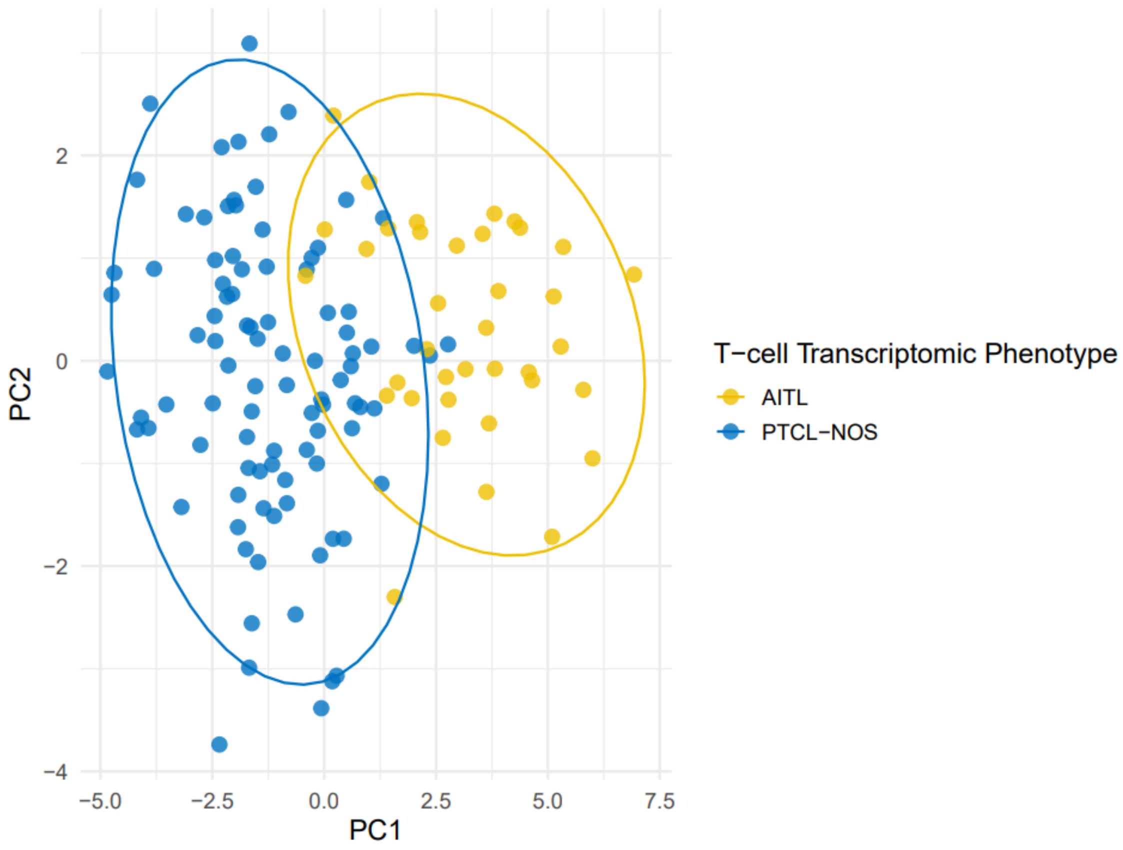
# Unsupervised signature validation

To empirically verify that our signature constructed a pure AITL space ready for the HL projection, we applied **Principal Component Analysis** (PCA). It computes orthogonal linear projections that maximize data variance by solving the eigenvalue problem on the data covariance matrix $\boldsymbol{\Sigma}$:

$$\boldsymbol{\Sigma}\mathbf{v} = \lambda\mathbf{v},$$

where eigenvectors $\mathbf{v}$ (principal components) corresponding to the largest eigenvalues $\lambda$ form the new axes.

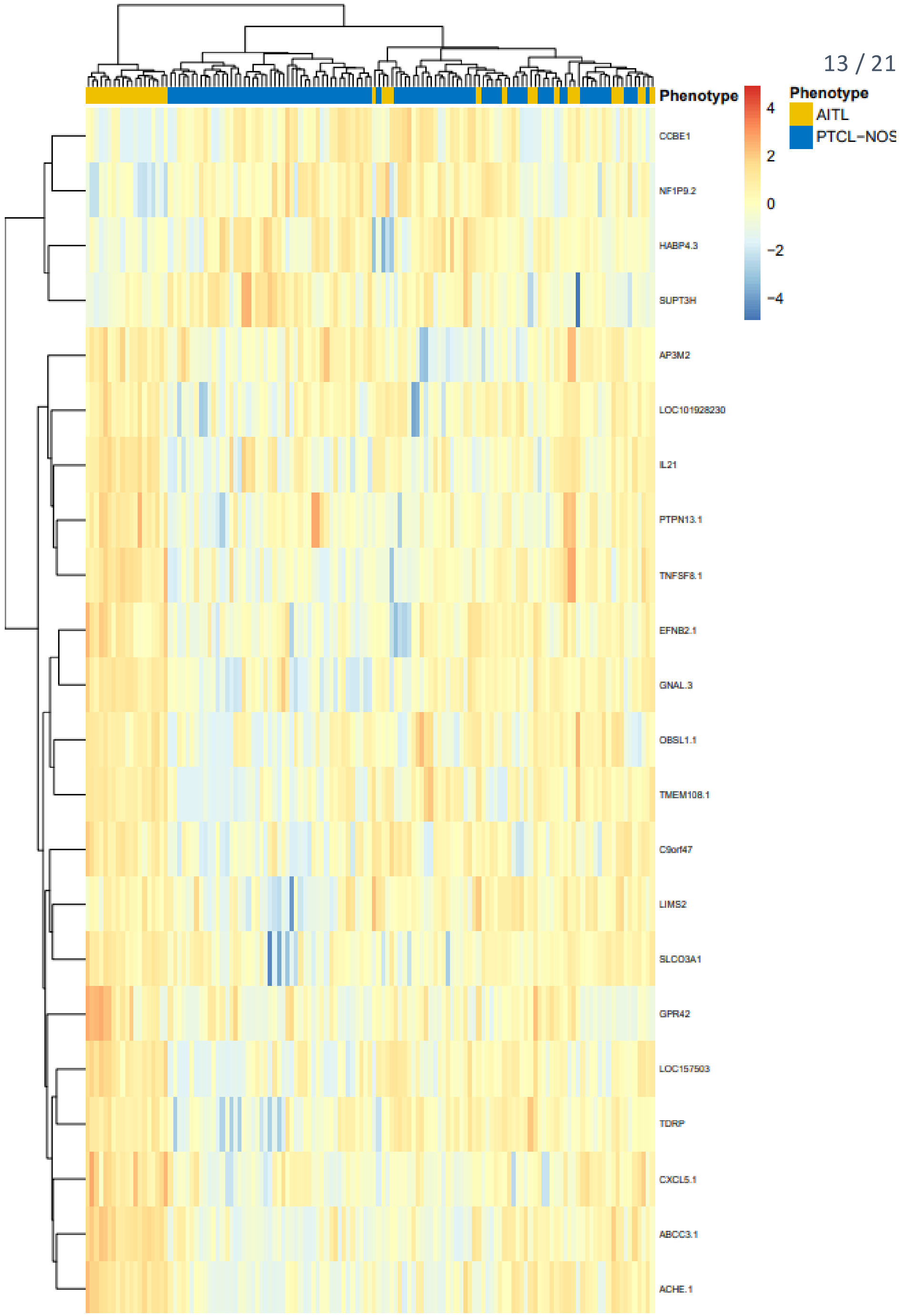
This confirms the selected genes naturally separate the phenotypes, except for a small overlap, defining the geometric space for our upcoming mimicry test.



# Unsupervised Transcriptomic Stratification

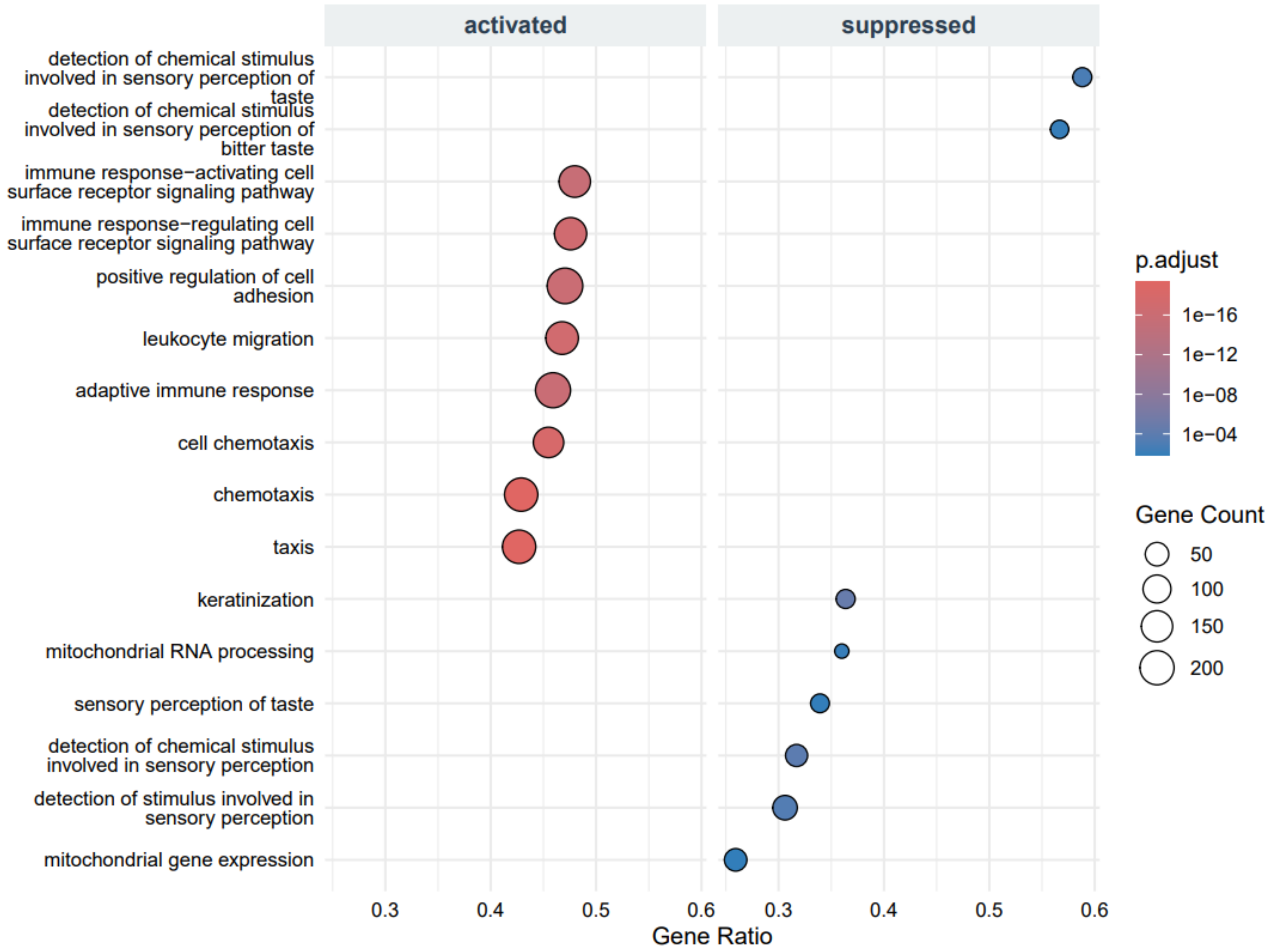
We concurrently mapped the underlying biological taxonomy via **Agglomerative Hierarchical Clustering**. This algorithm builds a structural dendrogram by recursively merging clusters $A$ and $B$ that minimize a specific linkage distance $d(A, B)$.

This mathematically reveals that our pipeline successfully isolated the pure AITL blueprint from the PTCL-NOS contrast background. Note that the signature stratifies the patients **based exclusively on the 22 transcripts**, without relying on clinical labels.



# GSEA reveals AITL mimicry biology

GSEA confirms that the 22-gene signature captures the microenvironmental shifts underlying AITL morphological mimicry.




The aggressive AITL phenotype is characterized by:

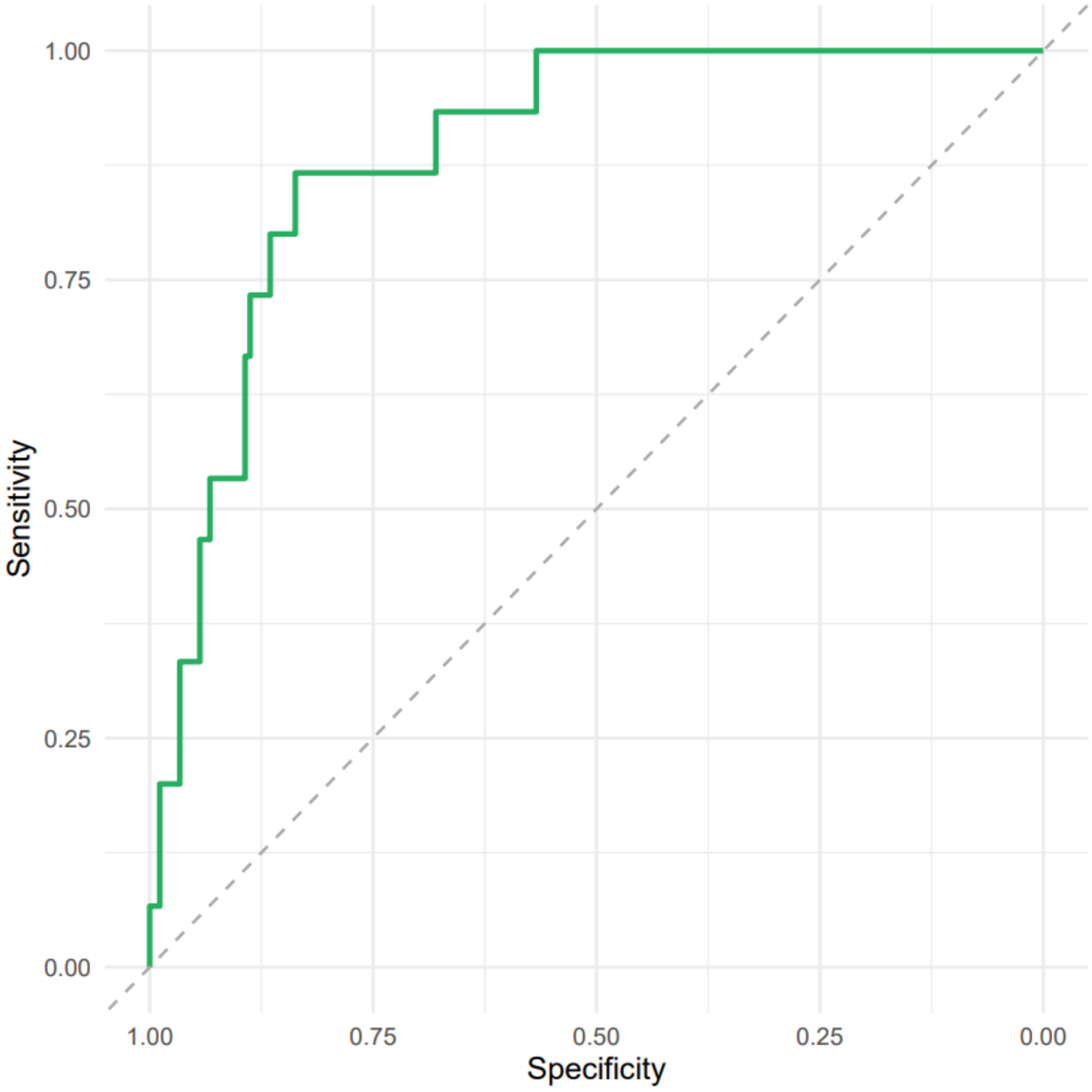
- suppression of cell adhesion
- reduced leukocyte migration
- hyper-activation of mitochondrial gene expression
- Activation of immune response
- Cell chemotaxis

Together, these changes suggest a profound remodeling of tissue architecture and immune-cell organization, generating the reactive inflammatory background that can make **AITL morphologically resemble classical Hodgkin Lymphoma**.

# Cross-cohort Robustness

- Our model achieved an **internal AUC** of **0.875** on the hold-out test set (data not used for training).
- To prove suppression of high-dimensional overfitting, the signature was projected onto an independent external cohort ($N = 193$) post-ComBat harmonization[a]. Performance increased to an **external AUC** of **0.890**, proving that our AITL detector generalizes well.

[a] To avoid bias driven by different mechanisms of data gathering.



# Diagnostic Validation

Table: Genomic Reclassification (Internal Hold-Out Set, $n = 26$)

| **True Clinical Label \ Predicted** | **AITL** | **PTCL-NOS** |
|---|---|---|
| **AITL** | 3 | 1 |
| **PTCL-NOS** | 1 | 21 |

Table: Genomic Reclassification (External Validation Cohort, $N = 193$)

| **True Clinical Label \ Predicted** | **AITL** | **PTCL-NOS** |
|---|---|---|
| **AITL** | 13 | 2 |
| **PTCL-NOS** | 49 | 129 |

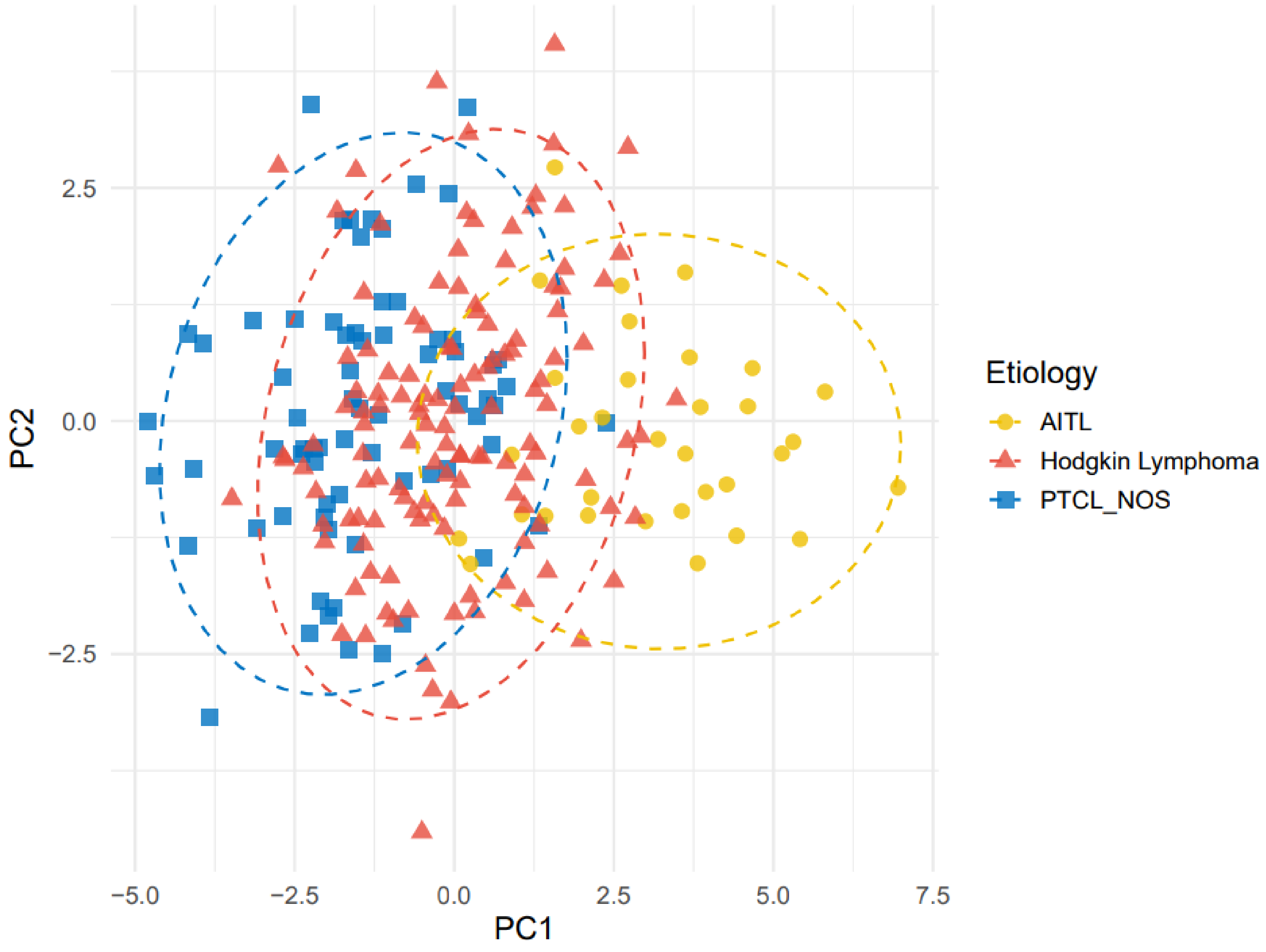
# Morphological Mimicry



With our pure AITL signature validated, we projected an independent cohort of officially diagnosed classical Hodgkin Lymphoma (GSE17920) into this space.

If standard pathology were infallible, HL would form an isolated cluster. Instead, they map almost entirely within the aggressive AITL/PTCL-NOS transcriptomic continuum.

This proves the **morphological mimicry** hypothesis. A substantial subset of rapid-onset secondary cases likely reflects an occult primary T-cell clone misdiagnosed at baseline due to overlapping microenvironments.



# Reconciling Epidemiology with Molecular Ground Truth



From our work, it emerges that the linear paradigm of iatrogenic mutagenesis for early sNHL is biologically incomplete.

First, our macro model proves systemic chemotherapy does not constitute an independent hazard, while it is **baseline severity** that **dictates outcomes**.

Second, our micro-genomic projection proves that **primary HL and aggressive T-cell neoplasms overlap deeply**.

Therefore, many rapid "secondary" T-cell lymphomas are not treatment-induced mutations, but the **clinical unmasking of an occult primary clone misdiagnosed at baseline due to overlapping cellular microenvironments**.

# What This Study Cannot Settle

*Honest scope of inference for the reported findings.*

01

## No granular dose data

SEER provides treatment as binary categorical variables. Cumulative dose-response effects (e.g. ABVD vs escalated BEACOPP) cannot be modelled, so Cox estimates are population-level marginal averages.

02

## Microarray-era platforms

The 22-gene signature was discovered and externally validated on legacy microarrays. ComBat harmonisation proves cross-cohort stability, but RNA-seq cross-platform calibration will be needed before any clinical assay.

03

## Retrospective design

Both the epidemiological and transcriptomic cohorts are retrospective. Causal inference is constrained, findings rest on high-confidence observational associations, not interventional evidence.

# Future Directions



**01**

## Diagnostic Paradigm Shift:

Our framework shows that standard localized morphology (H&E and limited IHC) is severely vulnerable to microenvironmental mimicry, mandating a reevaluation of diagnostic protocols.

**02**

## Translational Machine Learning:

Our molecular signature was derived and validated on historical microarray platforms rather than contemporary RNA-sequencing. While orthogonal ComBat harmonization proved the signature's stability and ability to generalize (external AUC of 0.890), translating this exact 22-gene panel into a routine clinical assay will require prospective cross-platform calibration.

**03**

## Precision Hematology:

Integrating non-invasive molecular tracking (e.g., circulating tumor DNA dynamics) and specialized T-cell clonality assays directly at the time of primary cHL diagnosis will allow to proactively catch the occult "mimicking" clones at Day 0.

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